Description of Teaching Interests

Every teaching or training role I have held shares one shape, whether the learner was a graduate student meeting an electron microscope for the first time, a ninth grader meeting acid-base chemistry for the first time, or a warehouse associate meeting a new shift-management system for the first time: build enough structure that someone who has never done the task can do it correctly and safely, then step back before the structure becomes a crutch. I did not learn this from a pedagogy course; I learned it because I have taught in nearly every register available to me — as a graduate mentor training peers on instrumentation, as a secondary-school teacher of record across four science subjects, as a paid and volunteer tutor, and as the person at Amazon who turned a skill that lived only in my own head into a repeatable training program for others. The approach I would bring to a biomedical engineering classroom or lab is the one I already run in each of those settings: start with the smallest safe unit of the task, make the standard of competence explicit and checkable, and hand off ownership as soon as the learner can meet that standard without me.

Evidence: Teaching and Instructional Design Across Levels

The most direct evidence comes from my doctoral training at UTEP's Inspired Materials & Stem Cell-Based Tissue Engineering Lab, where over four years I mentored more than thirty undergraduate and graduate students in electron microscopy technique and 3D bioprinting. That instruction was necessarily one-on-one and hands-on with expensive, easily-misused instrumentation, so it could only be taught the scaffolded, competency-checked way I still favor. In 2019 I also taught the electron-microscopy module within *BME 4371, Biomaterials Characterization and Biocompatibility* — I was the module instructor for that unit, not the instructor of record for the course as a whole — and its outcome is one I am glad to point to directly: it produced a co-authored, peer-reviewed publication, with students as collaborators on the data rather than passive recipients of a lecture. I see that as a small existence proof of exactly the kind of experiential, high-impact instruction this search values: a technique taught hands-on, inside a course, generating real scholarly output.

Before graduate school, I was the teacher of record for physics (classical mechanics through introductory quantum and solid-state physics), chemistry (inorganic through organic/biochemistry), AP Environmental/Geological Science, and biology at Radford School in El Paso — spanning remedial general science for seventh and eighth graders up to AP-level material, sometimes in the same week. That breadth is unusual for someone headed toward a biomedical engineering faculty position, and I consider it an asset: it forced me to diagnose, in real time, what a specific student actually did and did not understand, rather than teach to an assumed baseline. I supplemented that year with several years of paid and volunteer tutoring spanning algebra through differential equations, biochemistry, anatomy, corrosion, and mechanics of materials — re-deriving the same concept several different ways until one landed for that particular student.

Most recently, at Amazon I built a training pipeline for Process Assistants where none had existed, trained fifteen people through it (nine concurrently), and was later asked by my own manager to train the Operations Manager and several Area Managers on shift procedures. I read that as instructional design for adult learners in a high-stakes, error-costly environment — the same discipline I would bring to a lab-safety module or technique-based course unit, exercised outside a university.

Teaching Directions Connected to My Research Program

I propose to build a research program in manufacturing and quality science for neuropsychiatric and neurodegenerative cell-based therapeutics and disease models. Connected to that program, I would want to develop — not import, but design from my own experience — instruction in three areas, offered here as directions I am prepared to build, not as courses that currently exist at Georgia Tech: a course on biomaterials and cell-product characterization methods, extending directly from the zeta-potential, conductivity, and microscopy-based techniques that were the substance of both my dissertation and the BME 4371 module I taught; a course bridging quality systems and biomanufacturing — critical quality attributes, critical process parameters, and quality-by-design thinking applied to cell and gene therapy manufacturing — drawing on my cGMP iPSC reprogramming and CLIA/COLA laboratory-compliance background; and a course in computational and quantitative methods for biomedical engineering, covering the bioprocess data pipelines, biostatistical modeling, and computational structural-biology tools (molecular docking, transcriptomic analysis) I have used across my postdoctoral and independent research. Each grows out of something I have actually built or run.

Mentoring Philosophy

Georgia Tech's posting asks for faculty who can mentor a diverse student body toward graduate education and build an inclusive, welcoming environment, and I would rather answer with evidence than a pledge. The clearest evidence is scale: thirty-plus students mentored one-on-one in a research lab, fifteen employees trained through a program I built from nothing, and several years of tutoring aimed at students — many without other access to that kind of help — in subjects from algebra to differential equations. It also includes community work I have kept doing regardless of what else was on my plate: I co-founded Latinos MD, which organized fundraising and MCAT-preparation resources for pre-medical students who otherwise could not access them; I served as a science fair judge at a local high school across two academic years; and I ran STEM tutoring and mentoring for engineering undergraduates through Tau Beta Pi and Alpha Chi during my own graduate training. None of that was required of me. I intend to run a research group where the on-ramp into a first research project, and into applying to graduate school, is treated as seriously as the research question itself — the students who most need that on-ramp built explicitly are usually the ones who benefit most from having it, and I have spent a career building it in whatever setting I happened to be in.